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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Hostel Access Management System

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the functional and non-functional requirements for the Hostel Access Management System. The system is designed to manage and control access to university hostels by students, hostel administrators, and security personnel.

1.2 Scope

The Hostel Access Management System is a software application that controls and monitors entry and exit into university of Eswatini hostels. The system will:

- Register students and hostel residents.
- Authenticate users using student ID cards or biometric credentials.
- Monitor hostel entry and exit activities.
- Generate reports for hostel management.
- Improve hostel security and access control.
- Restrict unauthorized access.
- Maintain a database of residents and visitors.

The system will be used by hostel administrators, security officers, students, and visitors.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
Admin	Hostel Administrator
RFID	Radio Frequency Identification
User	Any authorized system user
Database	Organized collection of system data
Authentication	Verification of user identity
Access Log	Record of hostel entry and exit

1.4 References

- IEEE Std 830-1998 Software Requirements Specification Standard.
- University Hostel Regulations.
- Software Engineering Principles and Practices.

1.5 Overview

This document describes the overall system requirements, interfaces, functional requirements, process descriptions, data specifications, and data dictionary for the Hostel Access System.

2. Overall Description

2.1 Product Perspective

The Hostel Access System is a centralized web-based and database-driven system connected to hostel entry devices such as RFID scanners, biometric scanners, or barcode readers.

2.2 Product Functions

The system will perform the following functions:

- User registration.
- User login and authentication.
- Student hostel access control.
- Visitor registration.

- Entry and exit monitoring.
- Report generation.
- Notification and alerts.
- User account management.

2.3 User Classes and Characteristics

User Class	Description
Administrator	Manages users, hostel records, and reports
Security Officer	Monitors hostel access and verifies identities
Student	Accesses hostel facilities
Visitor	Temporary guest requesting access

2.4 Operating Environment

- Windows or Linux operating systems.
- Web browsers such as Chrome and Firefox.
- MySQL or PostgreSQL database.
- Internet or local network connectivity.

2.5 Design and Implementation Constraints

- The system must comply with university security policies.
- User data must be protected.
- The system must support multiple users simultaneously.
- The system must operate continuously.

2.6 Assumptions and Dependencies

- Students possess valid student identification cards.
- Internet/network connection is available.
- Hostel administrators have access permissions.
- The server remains operational.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system shall provide:

- Login page.
- Dashboard for administrators.
- Student registration forms.
- Visitor registration forms.
- Access monitoring screen.
- Report generation interface.
- Alert and notification messages.

3.1.2 Hardware Interfaces

The system shall interface with:

- RFID card readers.
- Barcode scanners.
- Biometric fingerprint scanners.
- Computers and servers.
- CCTV monitoring systems.

3.1.3 Software Interfaces

The system shall interface with:

- MySQL database management system.
- Web browsers.
- Operating systems.
- University student information systems.

3.1.4 Communications Interfaces

The system shall support:

- TCP/IP communication.
- Secure HTTPS communication.
- Local Area Network (LAN) connections.
- Email and SMS notifications.

3.2 Functional Requirements

3.2.1 Information Flows

3.2.1.1 Data Flow Diagram 1 – Student Registration

3.2.1.1.1 Data Entities

- Student
- Administrator
- Hostel Database

3.2.1.1.2 Pertinent Processes

- Capture student details.
- Validate registration information.
- Store student data in database.

3.2.1.1.3 Topology

The administrator enters student information into the system. The system validates the information and stores it in the hostel database.

3.2.1.2 Data Flow Diagram 2 – Hostel Access Authentication

3.2.1.2.1 Data Entities

- Student
- Security Officer
- Access Control System
- Database

3.2.1.2.2 Pertinent Processes

- Scan student ID or biometric data.
- Verify user credentials.
- Grant or deny access.
- Record entry or exit log.

3.2.1.2.3 Topology

The student scans their ID card or fingerprint at the hostel entrance. The system validates the credentials using the database and either grants or denies access.

3.2.1.3 Data Flow Diagram 3 – Visitor Management

3.2.1.3.1 Data Entities

- Visitor
- Security Officer
- Visitor Database

3.2.1.3.2 Pertinent Processes

- Register visitor details.
- Validate host information.
- Generate temporary access pass.
- Store visitor records.

3.2.1.3.3 Topology

Visitors provide identification information to the security officer. The system records visitor details and generates temporary access permission.

3.2.2 Process Descriptions

3.2.2.1 Process 1 – Student Registration

3.2.2.1.1 Input Data Entities

- Student name
- Student ID
- Hostel room number
- Contact information

3.2.2.1.2 Algorithm or Formula of Process

1. Administrator enters student details.
2. System validates information.
3. System checks for duplicate records.
4. System stores information in database.
5. Confirmation message is displayed.

3.2.2.1.3 Affected Data Entities

- Student records
- Hostel allocation records

3.2.2.2 Process 2 – Authentication and Access Control

3.2.2.2.1 Input Data Entities

- Student ID card
- Fingerprint data
- User credentials

3.2.2.2.2 Algorithm or Formula of Process

1. User scans ID or fingerprint.
2. System retrieves stored credentials.
3. System compares credentials.
4. If credentials are valid, access is granted.
5. If invalid, access is denied and alert generated.
6. Access log is updated.

3.2.2.2.3 Affected Data Entities

- Access logs
- Security reports
- Student records

3.2.2.3 Process 3 – Visitor Registration

3.2.2.3.1 Input Data Entities

- Visitor name
- National ID/passport
- Host student details
- Time of visit

3.2.2.3.2 Algorithm or Formula of Process

1. Security officer captures visitor details.
2. System verifies host student.
3. Visitor information is stored.
4. Temporary pass is issued.
5. Visitor exit time is recorded upon departure.

3.2.2.3.3 Affected Data Entities

- Visitor records
- Access logs
- Security records

3.2.2.4 Process 4 – Report Generation

3.2.2.4.1 Input Data Entities

- Access logs
- Student records
- Visitor records

3.2.2.4.2 Algorithm or Formula of Process

1. Administrator selects report type.
2. System retrieves required data.
3. System generates report.
4. Report is displayed or printed.

3.2.2.4.3 Affected Data Entities

- Management reports
- Security reports

3.2.3 Data Construct Specifications

3.2.3.1 Construct 1 – Student Record

3.2.3.1.1 Record Type

Student Database Record

3.2.3.1.2 Constituent Fields

- Student ID
- Full name
- Gender
- Contact number
- Hostel room number
- RFID card number
- Fingerprint template

3.2.3.2 Construct 2 – Access Log Record

3.2.3.2.1 Record Type

Access Monitoring Record

3.2.3.2.2 Constituent Fields

- Log ID

- Student ID
- Entry time
- Exit time
- Access status
- Security officer ID

3.2.3.3 Construct 3 – Visitor Record

3.2.3.3.1 Record Type

Visitor Database Record

3.2.3.3.2 Constituent Fields

- Visitor ID
- Visitor name
- National ID/passport
- Host student ID
- Check-in time
- Check-out time

3.2.4 Data Dictionary

3.2.4.1 Data Element 1 – Student ID

3.2.4.1.1 Name

Student ID

3.2.4.1.2 Representation

Numeric

3.2.4.1.3 Units/Format

Example: 202302047

3.2.4.1.4 Precision/Accuracy

Must be unique and valid.

3.2.4.1.5 Range

8–15 characters.

3.2.4.2 Data Element 2 – Hostel Room Number

3.2.4.2.1 Name

Hostel Room Number

3.2.4.2.2 Representation

Numeric/Alphanumeric

3.2.4.2.3 Units/Format

Example: B12

3.2.4.2.4 Precision/Accuracy

Must correspond to existing room.

3.2.4.2.5 Range

1–10 characters.

3.2.4.3 Data Element 3 – Access Time

3.2.4.3.1 Name

Access Time

3.2.4.3.2 Representation

Date and Time

3.2.4.3.3 Units/Format

YYYY-MM-DD HH:MM:SS

3.2.4.3.4 Precision/Accuracy

Must record exact entry and exit time.

3.2.4.3.5 Range

24-hour time format.

3.2.4.4 Data Element 4 – Visitor ID

3.2.4.4.1 Name

Visitor ID

3.2.4.4.2 Representation

Alphanumeric

3.2.4.4.3 Units/Format

Passport/National ID Number

3.2.4.4.4 Precision/Accuracy

Must match provided identification document.

3.2.4.4.5 Range

6–20 characters.

Non-Functional Requirements

Performance Requirements

- The system shall respond within 3 seconds.
- The system shall support at least 500 simultaneous users.

Security Requirements

- User passwords shall be encrypted.
- Only authorized users shall access administrative functions.
- All access attempts shall be logged.

Reliability Requirements

- The system shall have 99% availability.
- Backup copies of data shall be created daily.

Usability Requirements

- The interface shall be user-friendly.
- The system shall support easy navigation.

Maintainability Requirements

- The system shall support future updates.
- Source code shall be modular and documented.

4. Appendices

Appendix A – Proposed Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: PHP or Python
- Database: MySQL
- Hardware: RFID Scanner, Fingerprint Scanner

Appendix B – Future Enhancements

- Mobile application support.
- Facial recognition integration.
- SMS alerts to parents or guardians.
- Integration with university payment systems.

Conclusion

The Hostel Access Management System will improve hostel security, automate access management, and provide efficient monitoring of students and visitors. This SRS document defines the requirements needed for successful system development according to the IEEE 830-1998 standard.